

Exit Probability Distributions

The initial derivation by Eugene d'Eon and Krivanek establishes the exit probability density function (ePDF) and cumulative distribution function (eCDF) as:

$$\text{ePDF}(x, u) = \frac{e^{x+\frac{x}{u}}}{1 - e^x x E_1(x)}$$

$$\text{eCDF}(x, u) = \frac{-1 - u e^{x+\frac{x}{u}} + e^x x E_1(x) - e^x x E_1\left(-\frac{x}{u}\right)}{-1 + e^x x E_1(x)}$$

Variables & Functions:

- $u \in [-1, 0]$: Cosine of the exit direction angle relative to the surface normal.
 - $x > 0$: Optical depth within the medium.
 - $E_1(z) = \Gamma(0, z) = \int_1^\infty \frac{e^{-zt}}{t} dt$: Exponential integral function.
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Weight Correction Derivation

When substituting standard directional sampling with the exit-resampled distribution $\text{ePDF}(x, u)$, an importance sampling weight correction factor $W(x, u)$ must be applied to every resampled ray to maintain an unbiased Monte Carlo estimator.

1. Original Directional Distribution:

When sampling directions according to the base boundary distribution, the base PDF with respect to solid angle measure is $p_{\text{orig}}(u)$.

2. Exit-Resampled Distribution:

The resampled distribution generates direction cosines $u \in [-1, 0]$ according to:

$$p_{\text{exit}}(u) = \text{ePDF}(x, u)$$

3. Weight Correction Factor:

The Monte Carlo weight adjustment factor is the ratio of the original target distribution to the sampling PDF:

$$W(x, u) = \frac{p_{\text{orig}}(u)}{\text{ePDF}(x, u)} = p_{\text{orig}}(u) \cdot \left[\frac{1 - e^x x E_1(x)}{e^{x+\frac{x}{u}}} \right]$$

Simplifying the exponential terms gives:

$$W(x, u) = p_{\text{orig}}(u) \cdot (1 - e^x x E_1(x)) e^{-x(1 + \frac{1}{u})}$$

Implementation Details

- **Inversion Method:** Since eCDF^{-1} lacks a closed-form analytic expression, sample directions are drawn by solving $\text{eCDF}(x, u) - \xi = 0$ using **Newton-Raphson root finding** initialized at $u_0 = -1$.
- **Exponential Integral Approximation:** For efficient evaluation in C++/HLSL shaders, the piecewise rational approximations from *Abramowitz & Stegun* (§5.1.53 / §5.1.56) maintain a relative accuracy within 10^{-7} to 10^{-6} across the full numerical domain.